

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 8 — SQL Joins — Part 1

Company Database (INNER, LEFT, RIGHT, FULL OUTER Joins)

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Roll No	2024-SE-11
Semester	04
Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	08
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Master table navigation using Primary Key (PK) and Foreign Key (FK) constraints.
- Differentiate the behavior of INNER JOIN, LEFT JOIN, and RIGHT JOIN.
- Implement the Anti-Join Pattern (LEFT JOIN ... WHERE right_key IS NULL) to find unlinked records.
- Emulate FULL OUTER JOIN in MySQL using LEFT JOIN, RIGHT JOIN, and UNION.
- Avoid Cartesian Products by correctly scoping join predicates.

2. Database Structure

The database (joins_lab) models a small company with 4 related tables.

2.1 Tables

Table	Purpose
Department	Department name, location, and budget
Employee	Employee details, including a self-referencing ManagerID
Project	Projects, each linked to an owning department
Assignment	Junction table linking employees to projects, with weekly hours

3. Key Tasks / Concepts Implemented

- Task A1-A2: Mapping employees to departments (INNER JOIN vs. LEFT JOIN to keep unassigned employees).
- Task A3-A4: Departments with zero employees, and projects with no assigned department.
- Task A5-A6: Anti-join pattern — 'benched' employees with no project, and dormant projects with no staff.
- Task A7-A8: Filtered and sorted INNER JOIN queries (by department, by location).
- Task A9: Headcount per department using LEFT JOIN + COUNT.
- Task A10: Emulating a FULL OUTER JOIN using LEFT JOIN + RIGHT JOIN + UNION.

4. Implementation (phpMyAdmin)

All 10 join tasks were implemented and executed using phpMyAdmin under XAMPP.

Server: 127.0.0.1 » Database: joins_lab

Structure SQL Search Query Export Import Operations Privileges Routines Events

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ assignment	Browse Structure Search Insert Empty Drop	9	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ department	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ employee	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ project	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	32.0 KiB	-
4 tables	Sum	30	InnoDB	utf8mb4_general_ci	128.0 KiB	0 B

☐ Check all With selected:

Structure tab showing all 4 tables

SELECT e.EmpID, e.EmpName, d.DeptName, d.Location **FROM** Employee e **INNER JOIN** Department d **ON** e.DeptID = d.DeptID;

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

EmpID	EmpName	DeptName	Location
101	Ali Khan	Engineering	Lahore
102	Sara Iqbal	Engineering	Lahore
103	Hamza Raza	Engineering	Lahore
104	Ayesha Noor	Marketing	Karachi
105	Bilal Ahmed	Marketing	Karachi
106	Fatima Sheikh	Finance	Islamabad
107	Usman Tariq	Finance	Islamabad
108	Maira Javed	Research	Lahore
109	Zain Abbas	Research	Lahore
110	Nida Yousaf	Research	Lahore

INNER join query

✓ Showing rows 0 - 9 (10 total, Query took 0.0003 seconds.)

```
SELECT e.EmpID, e.EmpName, d.DeptName, d.Location FROM Employee e LEFT JOIN Department d ON e.DeptID = d.DeptID;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

EmpID	EmpName	DeptName	Location
101	Ali Khan	Engineering	Lahore
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103	Hamza Raza	Engineering	Lahore
104	Ayesha Noor	Marketing	Karachi
105	Bilal Ahmed	Marketing	Karachi
106	Fatima Sheikh	Finance	Islamabad
107	Usman Tariq	Finance	Islamabad
108	Maira Javed	Research	Lahore
109	Zain Abbas	Research	Lahore
110	Nida Yousaf	Research	Lahore

LEFT join query

Structure SQL Search Query Export Import Operations Privileges Routines Events M

```
SELECT e.EmpID, e.EmpName, d.DeptID, d.DeptName, d.Location FROM Employee e LEFT JOIN Department d ON e.DeptID = d.DeptID UNION SELECT e.EmpID, e.EmpName, d.DeptID, d.DeptName, d.Location FROM Employee e RIGHT JOIN Department d ON e.DeptID = d.DeptID;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

EmpID	EmpName	DeptID	DeptName	Location
101	Ali Khan	10	Engineering	Lahore
102	Sara Iqbal	10	Engineering	Lahore
103	Hamza Raza	10	Engineering	Lahore
104	Ayesha Noor	20	Marketing	Karachi
105	Bilal Ahmed	20	Marketing	Karachi
106	Fatima Sheikh	30	Finance	Islamabad
107	Usman Tariq	30	Finance	Islamabad
108	Maira Javed	40	Research	Lahore
109	Zain Abbas	40	Research	Lahore
110	Nida Yousaf	40	Research	Lahore
NULL	NULL	50	Sales	Karachi

FULL OUTER JOIN

5. Conclusion

This lab built a solid foundation in relational joins, including the anti-join pattern and emulating FULL OUTER JOIN, which MySQL does not support natively.